GlobPOP: A 31-year (1990-2020) global gridded population dataset generated by cluster analysis and statistical learning

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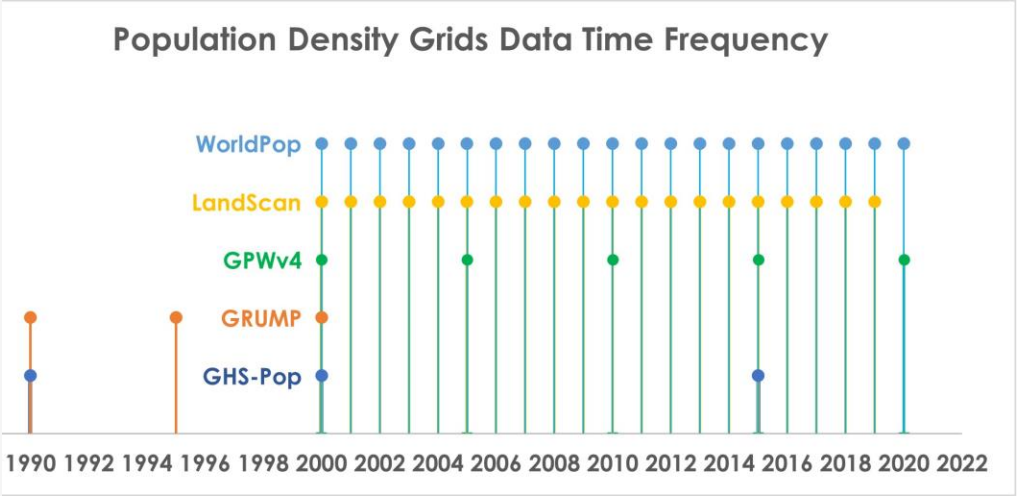
1 Introduction

1.1 Challenges in Gridded Population Data

The map we need if we want to achieve 'leaving no one behind' target

Source: online at OurWorldInData.org

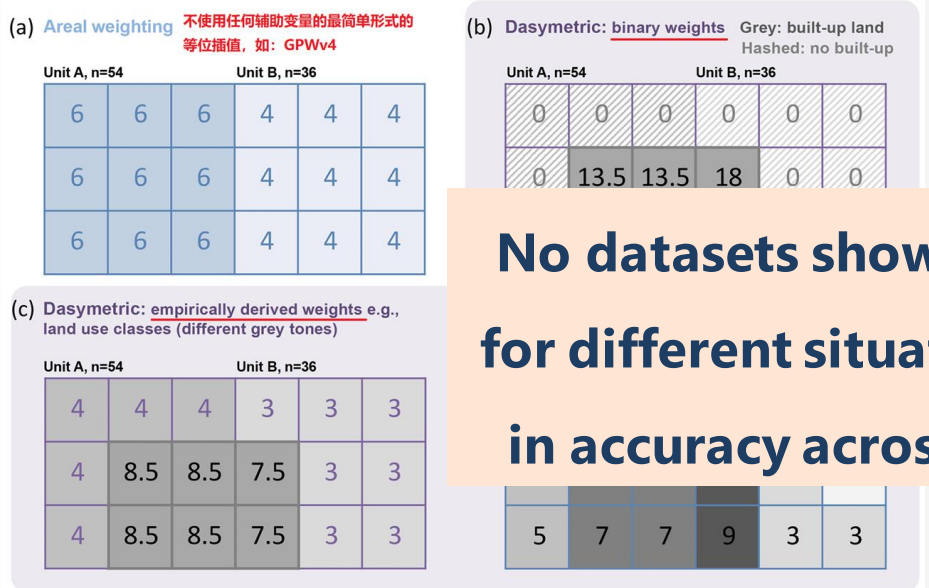
No continuous long-term gridded population dataset before 2000



Specifically, of the 17 Sustainable Development Goals and 169 targets set by the United Nations in 2015, approximately half of the indicators require accurate and spatially explicit demographic data

1 Introduction *1.2 Navigating Accuracy Dilemmas*

Common sources of uncertainty: methodology and ancillary data



Top-down method

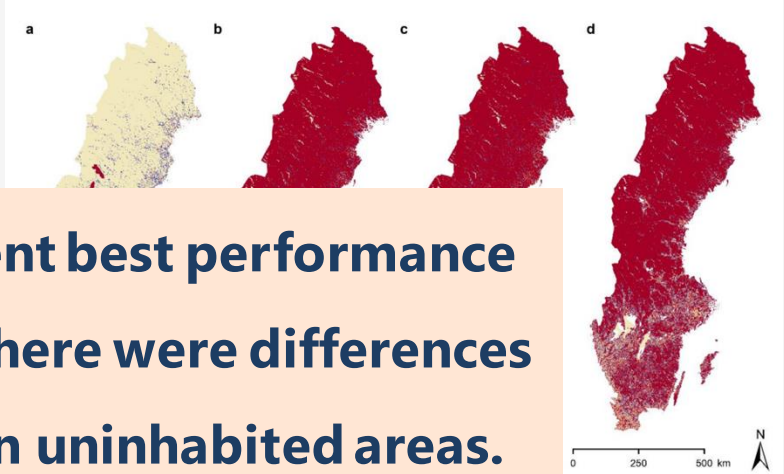


Fig. A1 Relative differences between candidate populations and the known population.

(a) GHS-POP 2015, (b) GPWv4 2015, (c) GRUMP 2000, (d) WorldPop 2015. GPWv4

Archila Bustos et al. , 2020

No datasets showed consistent best performance for different situations, and there were differences in accuracy across datasets in uninhabited areas.

The reliability and uncertainty of population data products are typically described in **documentation or validated in specific countries and regions.**

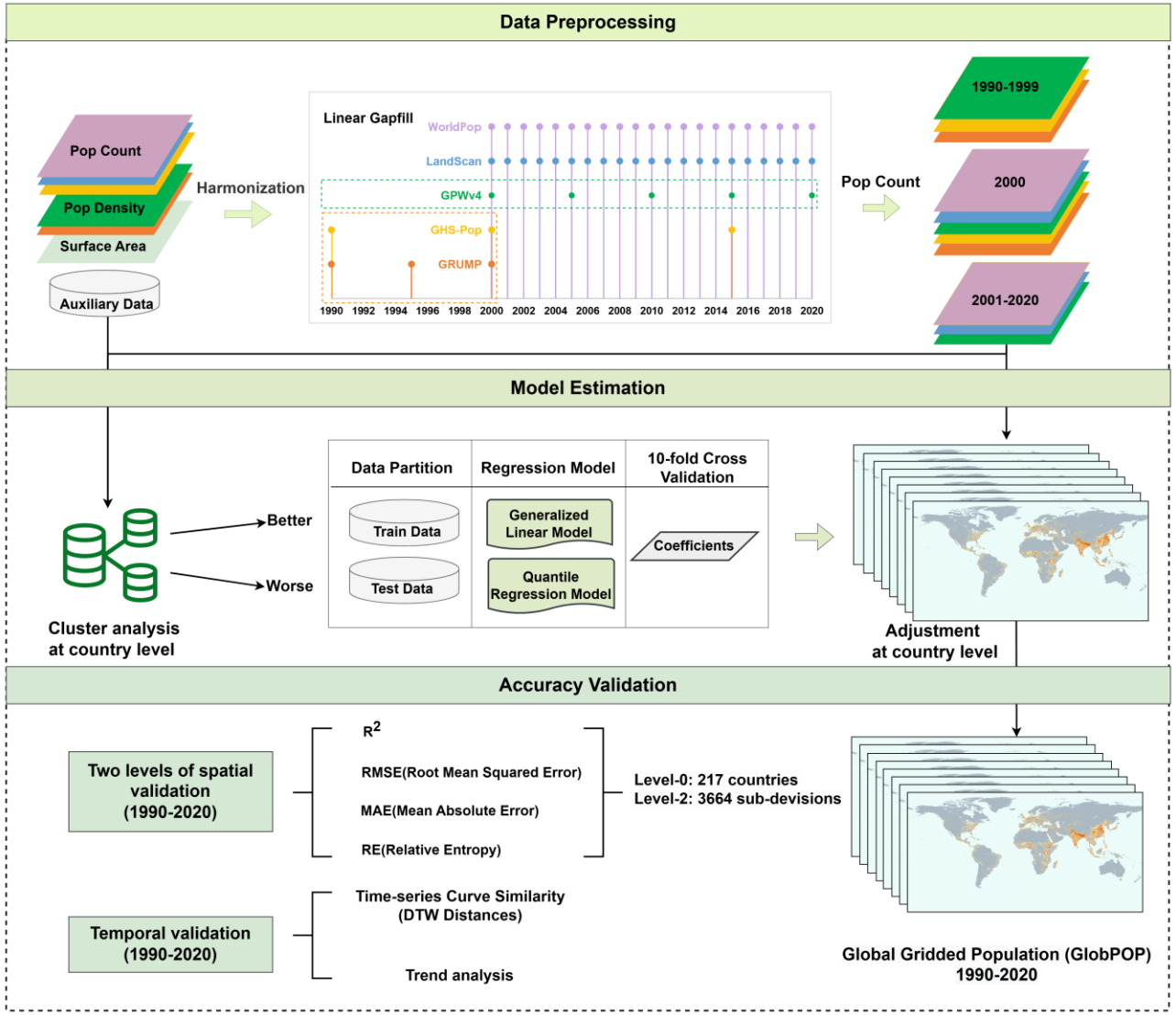
[1] Archila Bustos et al. A pixel level evaluation of five multitemporal global gridded population datasets: a case study in Sweden, 1990–2015. *Popul Environ* **42,** 255–277 (2020).

1 Introduction

1.3 Global Insights and Innovative Solutions

? **Whether it is possible to integrate** these five sets of multi-source demographic data and **leverage the strengths of each data**

This study proposed a **data fusion framework to generate a continuous global gridded population (GlobPOP)** from 1990 to 2020 using the five existing products.



2 Methods

2.1 Materials

Dataset	Unit	Input population source	Resolution (arc-seconds)	Year	Coordinate System
GHS-POP	person/pixel	GPWv4.10	30″	1990,2000	WGS-84
GRUMP	person /km^2	GPWv3	30″	1990,1995, 2000	WGS-84
GPWv4.11	person /km^2	Census	30″	2000,2005,2010,2015,2020	WGS-84
LandScan	person / pixel	Census	30″	2000-2020	WGS-84
WorldPop	person / pixel	GPWv4 and Census	30″	2000-2020	WGS-84

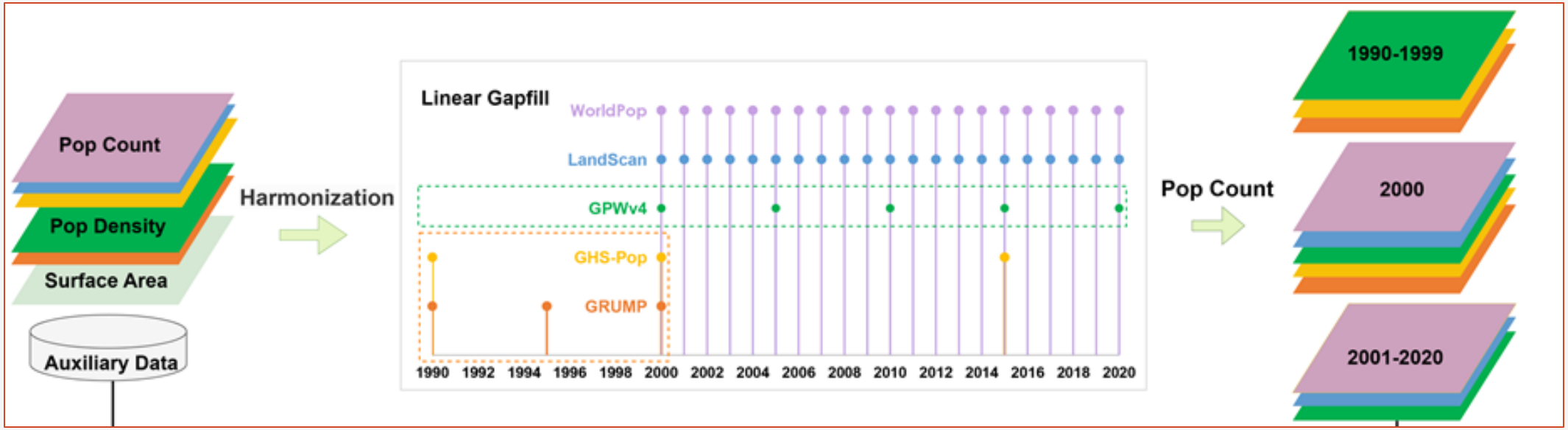
Detailed information of original input population data sources

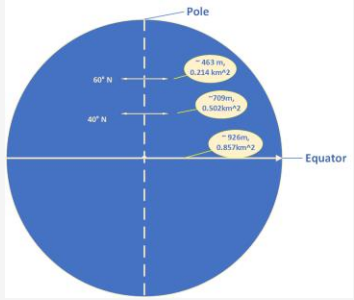
Data	Scale	Year
GDAM Shapefile	-	2018
Surface Area: GPWv4 land_water_area.tif	30″	2019
World Pop Prospect	217 Country Administration	2022
India Census	Level-2 Administration	1991,2001,2011
China Census	Level-2 Administration	2000,2010,2020
Angola Census	Level-2 Administration	2014
South Africa Census	Level-2 Administration	1996,2001,2011,2016
Nigeria Census	Level-2 Administration	1991,2006
Vanuatu Census	Level-2 Administration	2009,2016
New Zealand Census	Level-2 Administration	1996,2001,2006
UK Census	Level-2 Administration	1991-2020
US Census	Level-2 Administration	1990-2020

Auxiliary data

2 Methods

2.2 Data Preprocessing





$$y = y_1 + (y_2 - y_1) \cdot \frac{t - t_1}{t_2 - t_1}$$

Data harmonization: raster data conversions and census data regulations (217 countries)

Linear gapfiling process: linear interpolation and extrapolation at the pixel level.

2 Methods

2.3 Model Estimation - Cluster Analysis for spatial consistency

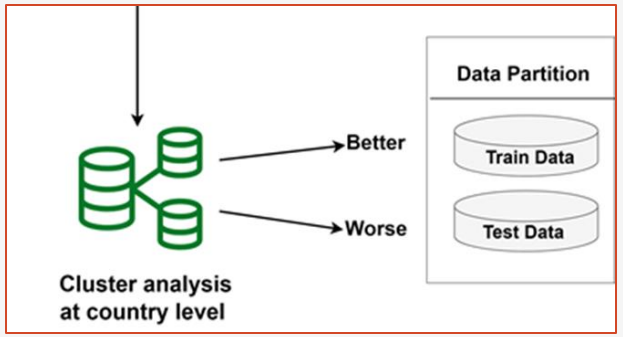
? Why cluster analysis

➢ **K-Means cluster analysis:**

1. Four metrics for quantifying similarity.

2. Identification of homogeneous groups

3. Significance for population estimation

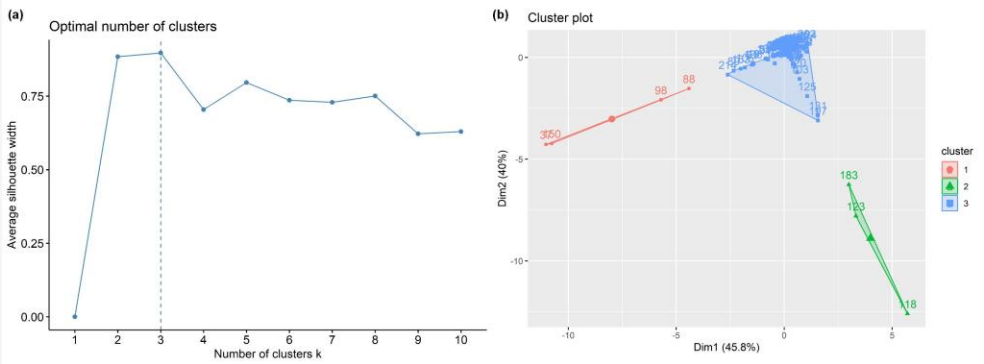


$$\text{APE (Absolute Percentage Error)} = \frac{X_i - Y_i}{X_i}$$

$$\text{SE (Squared Error)} = (X_i - Y_i)^2$$

$$\text{SLE (Squared Logarithmic Error)} = (\ln(1+X_i) - \ln(1+Y_i))^2$$

$$\text{Dif (Difference)} = (Y_i - X_i)$$



Year	Grump	GPW	GHS	LandScan	WorldPop
2000	India	India, Maldives, Netherlands Antilles, Guadeloupe	India, Pakistan, Netherlands Antilles, Maldives, Guadeloupe	India	India

Example: worse case

2 Methods

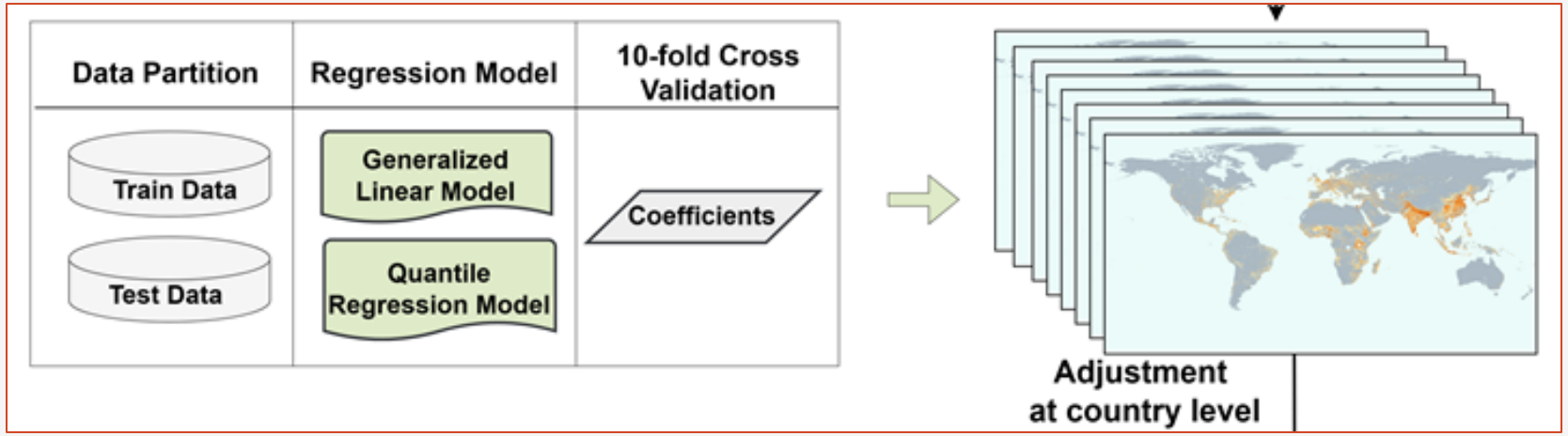
2.3 Model Estimation - Regression Models for Fusion

Generalized linear model (GLM)

Quantile regression model (QRM)

$$Y_t = a_{1,t}X_{1,t} + a_{2,t}X_{2,t} + \cdots + a_{n,t}X_{n,t} + \varepsilon_t$$

non-normal distribution, robust to outliers, control the effects of the confounding factors



- L-BFGS-B algorithm for non-negativity constraint.
- 10-fold cross-validation.
- 200 iterations.
- Population adjustment.

2.4 GlobPOP Validation - Spatial and Temporal

Accuracy indexes:

Metrics	Equation	Description
R^2	$R^2 = 1 - \frac{\sum_{i=1}^{n}(x_i - y_i)^2}{\sum_{i=1}^{n}(\bar{x} - x_i)^2}$	Higher values indicate a better fit.
RMSE	$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - y_i)^2}$	Lower values indicate a better fit.
MAE	$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}\lvert x_i - y_i \rvert$	Lower values indicate a better fit.
RE	$RE = \int_{-\infty}^{\infty} P(x) \cdot \log(\frac{P(x)}{Q(y)})dx$	Higher values indicate a better fit.
DTW distance	Warping curve: $\phi(t) = \left(\phi_x(t), \phi_y(t)\right)$, t = 1,...,T $d_\phi(X,Y) = \sum_{t=1}^{T} d\left(\phi_x(t), \phi_y(t)\right) m_\phi(t)/M_\phi$ DTW distance: $D(X,Y) = \min d_\phi(X,Y)$	Lower values indicate a better fit.

Spatial validation

- Level-0: 217 countries
- Level-2 : 3664 sub-divisions

Temporal validation

- Dynamic Time Warping (DTW) for measuring time-series curve similarities
- Sen's slope estimator and non-parametric Mann-Kendall test for long time-series trend analysis

3 Results

3.1 Cluster Analysis Results

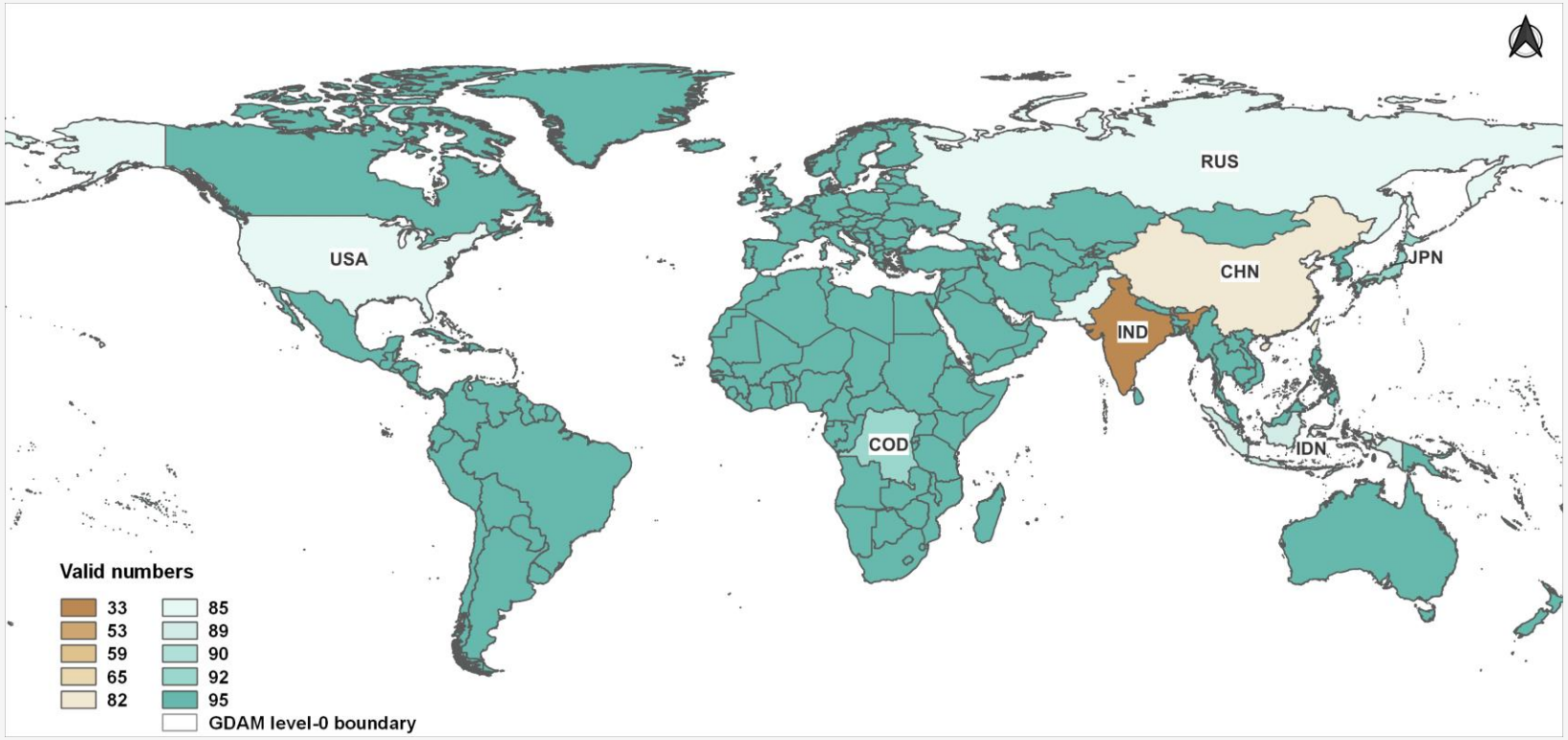


Fig.1 Accountability Distribution

- The uneven distribution of valid data sets highlights **that the reliability of these products fluctuates over time and is not uniform across all regions.**
- **Top 3 Countries with Low Accountability:** India, Guadeloupe, and the Republic of Maldives

3 Results

3.2 Spatial Accuracy Validation - Level-0

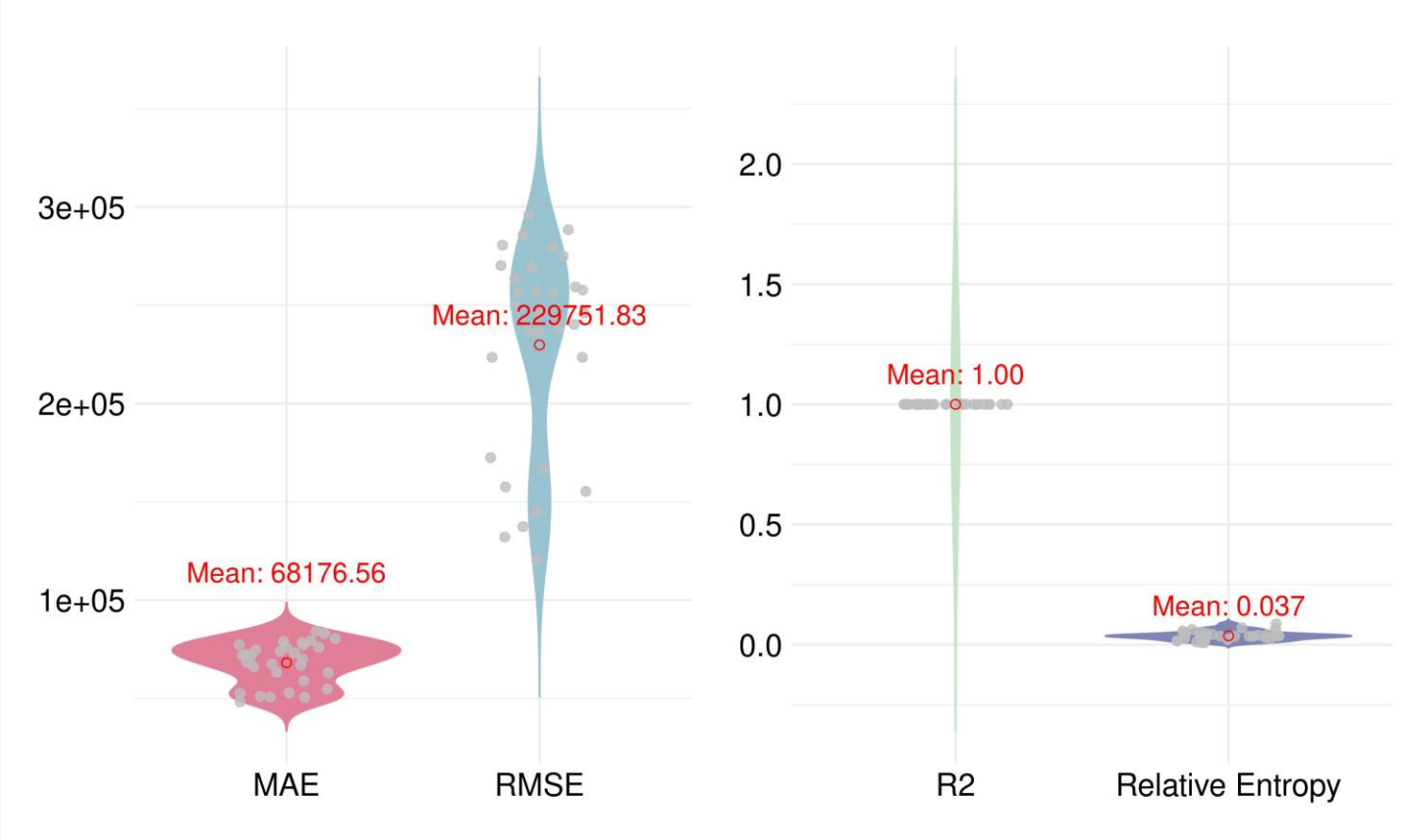


Fig.2 Accuracy metrics at the level-0 scale from 1990 to 2020

- The **QRM(quantile regression model)** was selected as the final population prediction model
- **GlobPOP: High accuracy in predicting country-level population estimation.**

3 Results

3.2 Spatial Accuracy Validation - Level-2

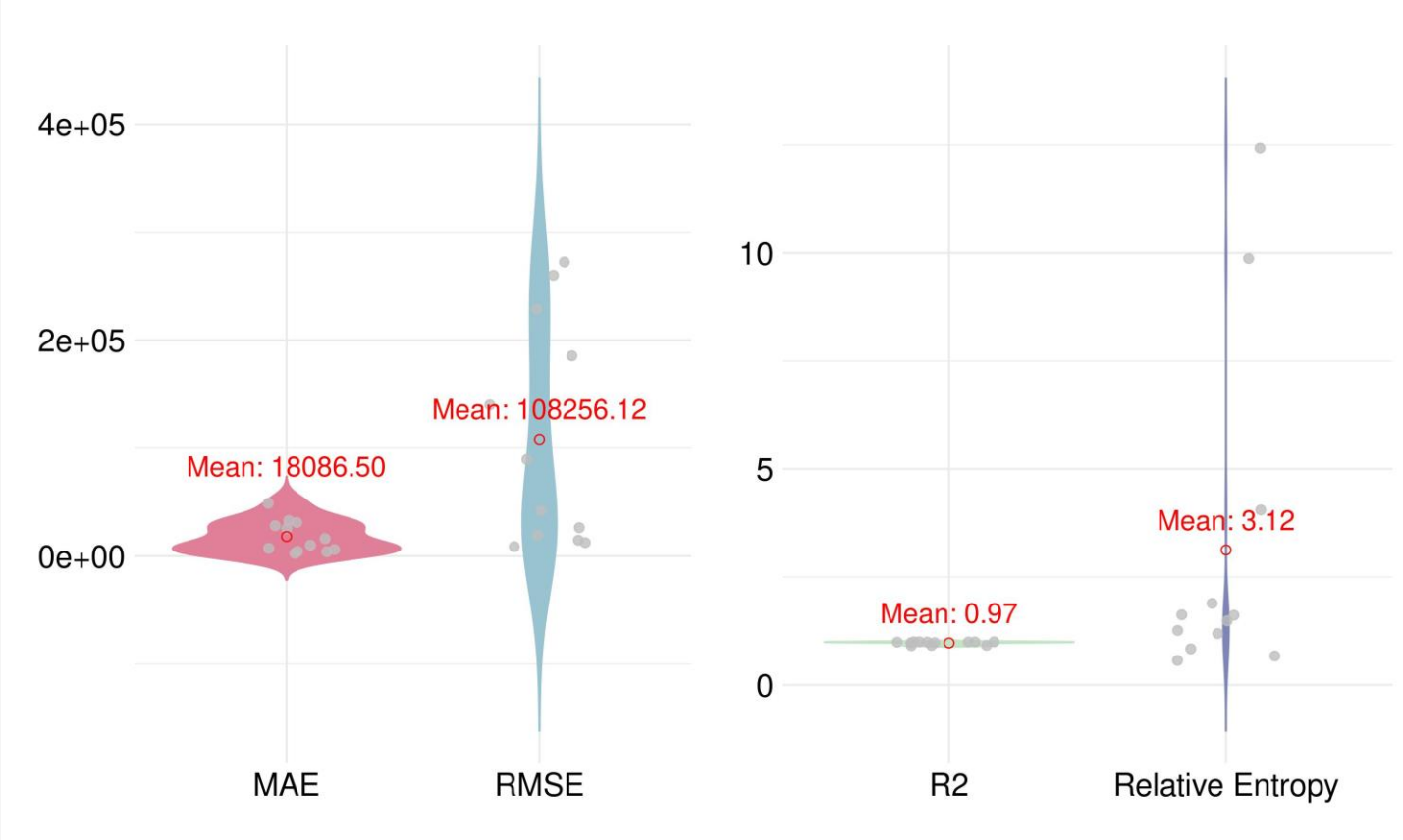


Fig.3 Accuracy metrics at the level-2 scale from 1990 to 2020

- The average R^2 is higher than 0.972 for all census available countries at the level-2 scale(Number: 3664).
- **GlobPOP: Good performance at the level-2 scale**

3 Results

3.3 Temporal Accuracy Validation - Level-0

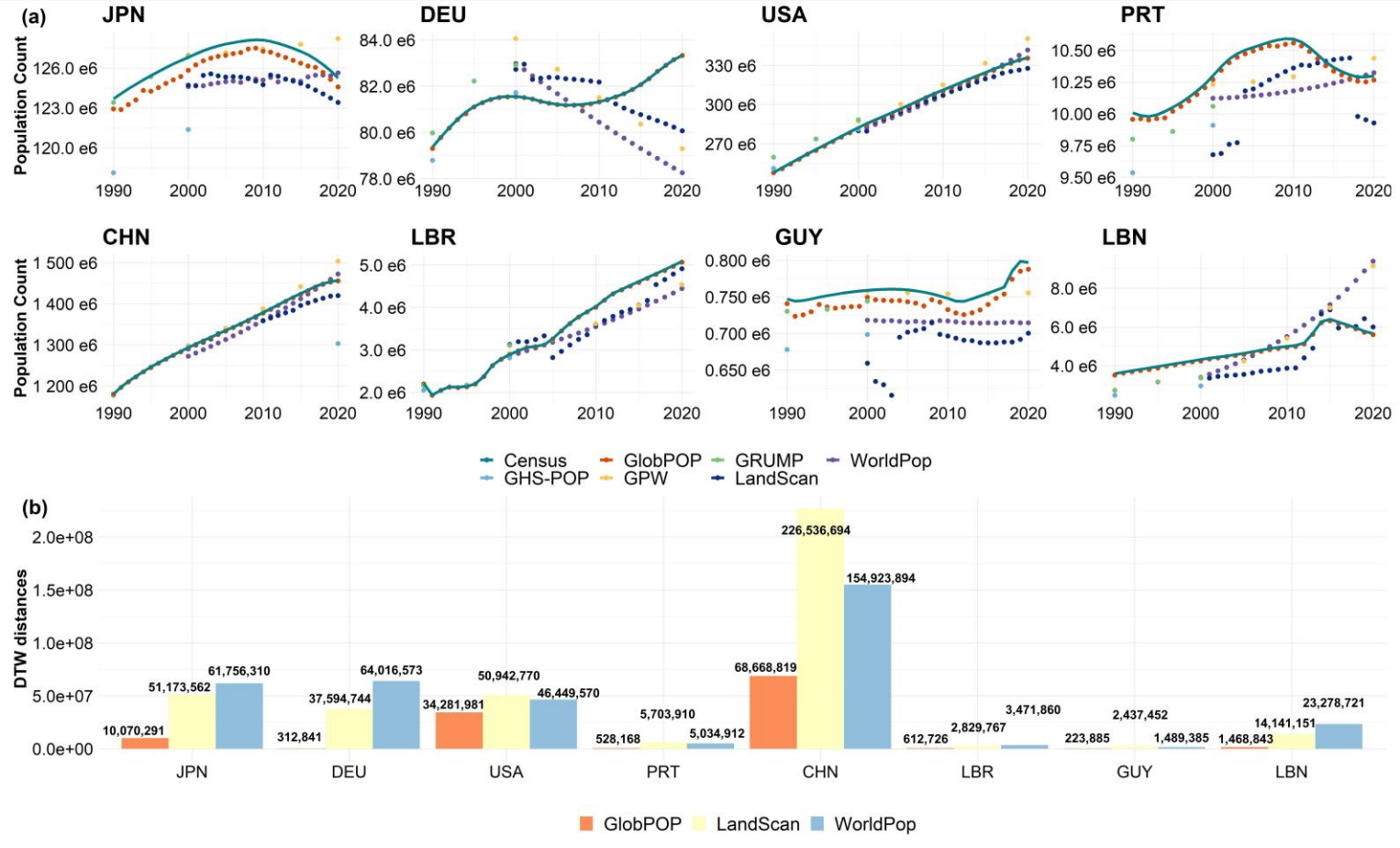


Fig.4 Comparison of the GlobPOP and the other datasets over the eight countries

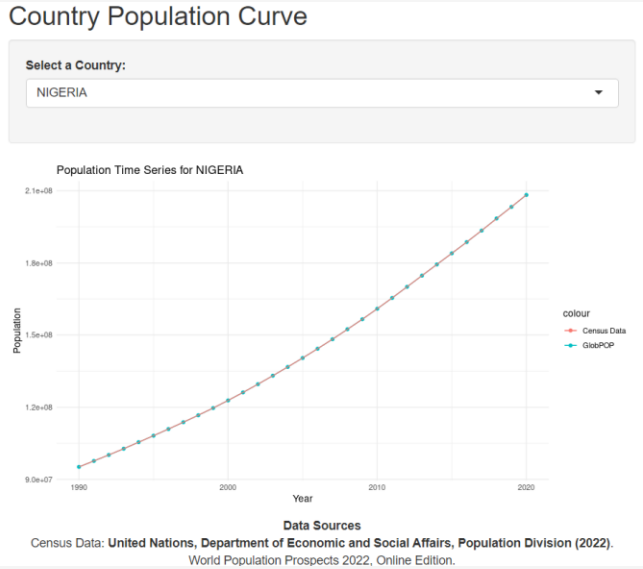


Fig.5 Population Curve Web application (217 countries).
https://globpop.shinyapps.io/GlobPOP/

- GlobPOP dataset shows the most consistent curve variations compared with the census curve

3 Results

3.3 Temporal Accuracy Validation – city level

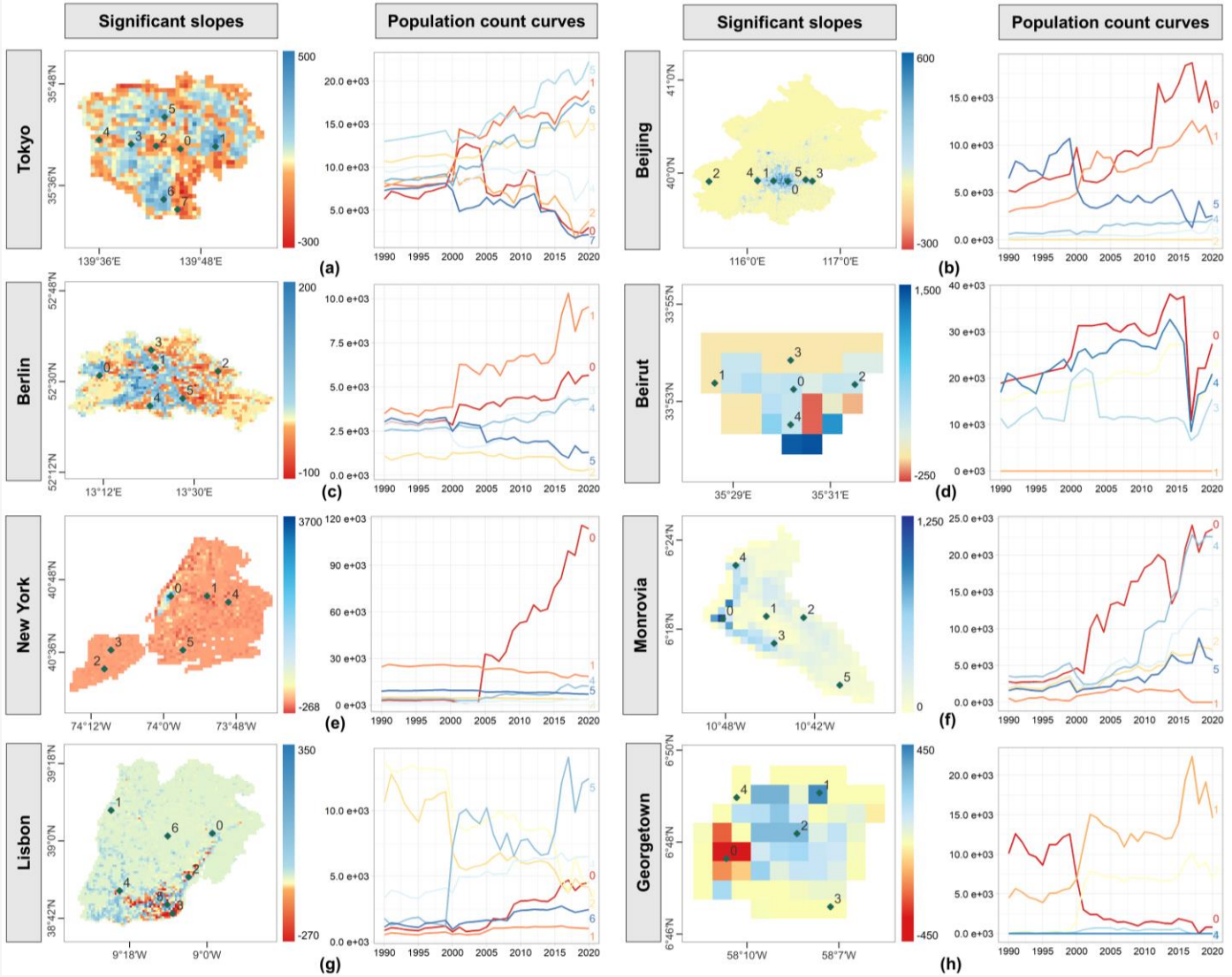


Fig.6 The temporal population trend analysis with significant slopes and pixel population curve variations in eight cities. (a) Tokyo in Japan; (b) Beijing in China; (c) Berlin in German; (d) Beirut in the Lebanese Republic; (e) New York in the United States; (f) Monrovia in Liberia; (g) Lisbon in Portugal; (h) Georgetown in Guyana.

- **GlobPOP: Captures population dynamics at the local scale**

3 Results

3.4 Spatial Evolution of Global Population (1990-2020)

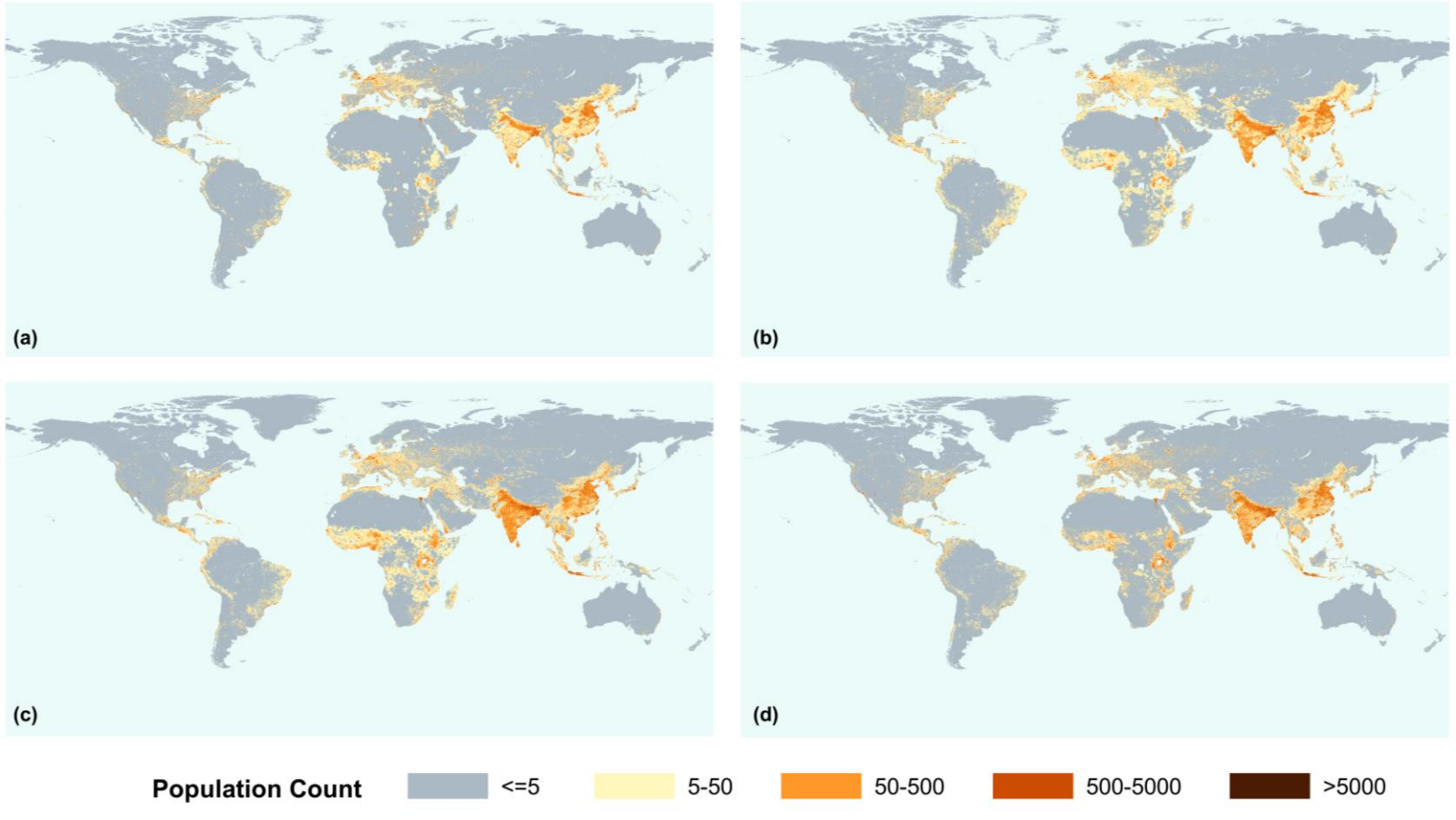


Fig.7 The global gridded population distribution from 1990 to 2020. (a) 1990; (b 2000; (c) 2010; (d) 2020.

- **GlobPOP: Understands the dynamic changes in global population**

4 Discussion

Benchmark test (Year 2000)

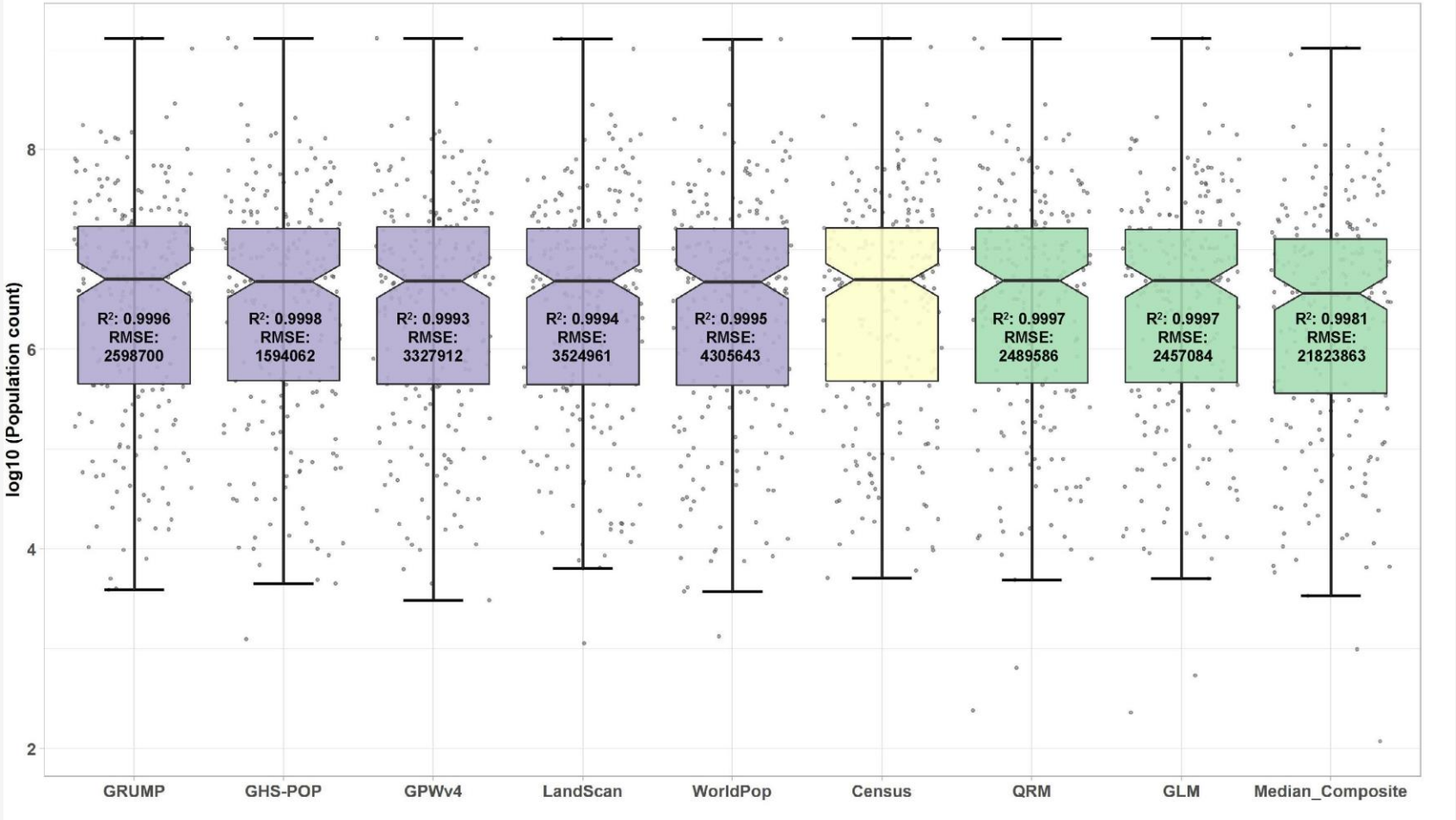


Fig. 8 Level-0 population count notched boxplots with data points after log10 transformation, and accuracy comparisons

4 Discussion

Benchmark test (Year 2000)

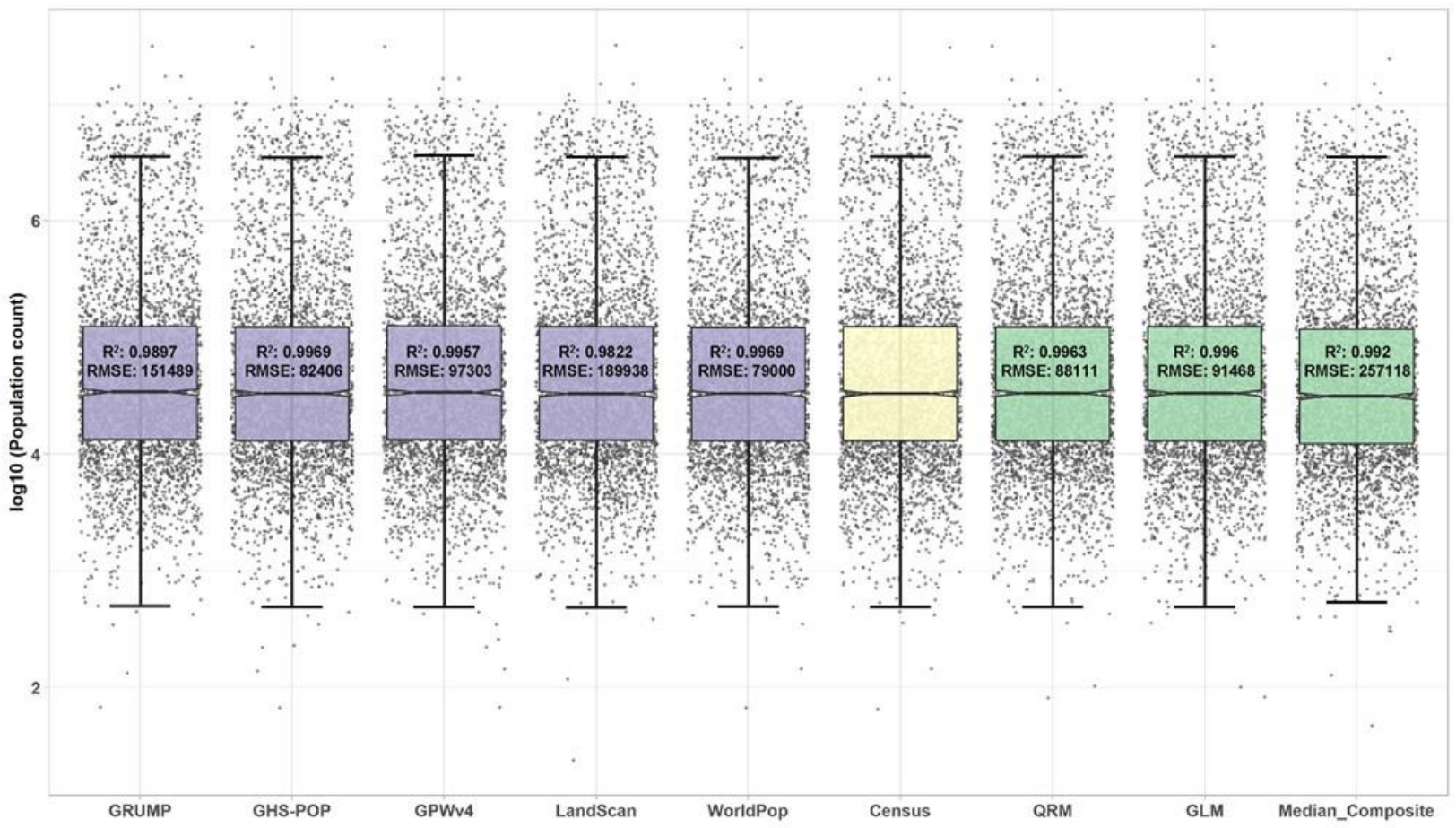


Fig. 9 Level-2 population count notched boxplots with data points after log10 transformation, and accuracy comparisons

? Why select QRM

- Consistent high accuracy

R^2:

Level-0 (0.9997)

Level-2 (0.9963)

- Superior performance at finer scale

4 Discussion

Pixel cases – visual inspection in rarely populated land cover areas

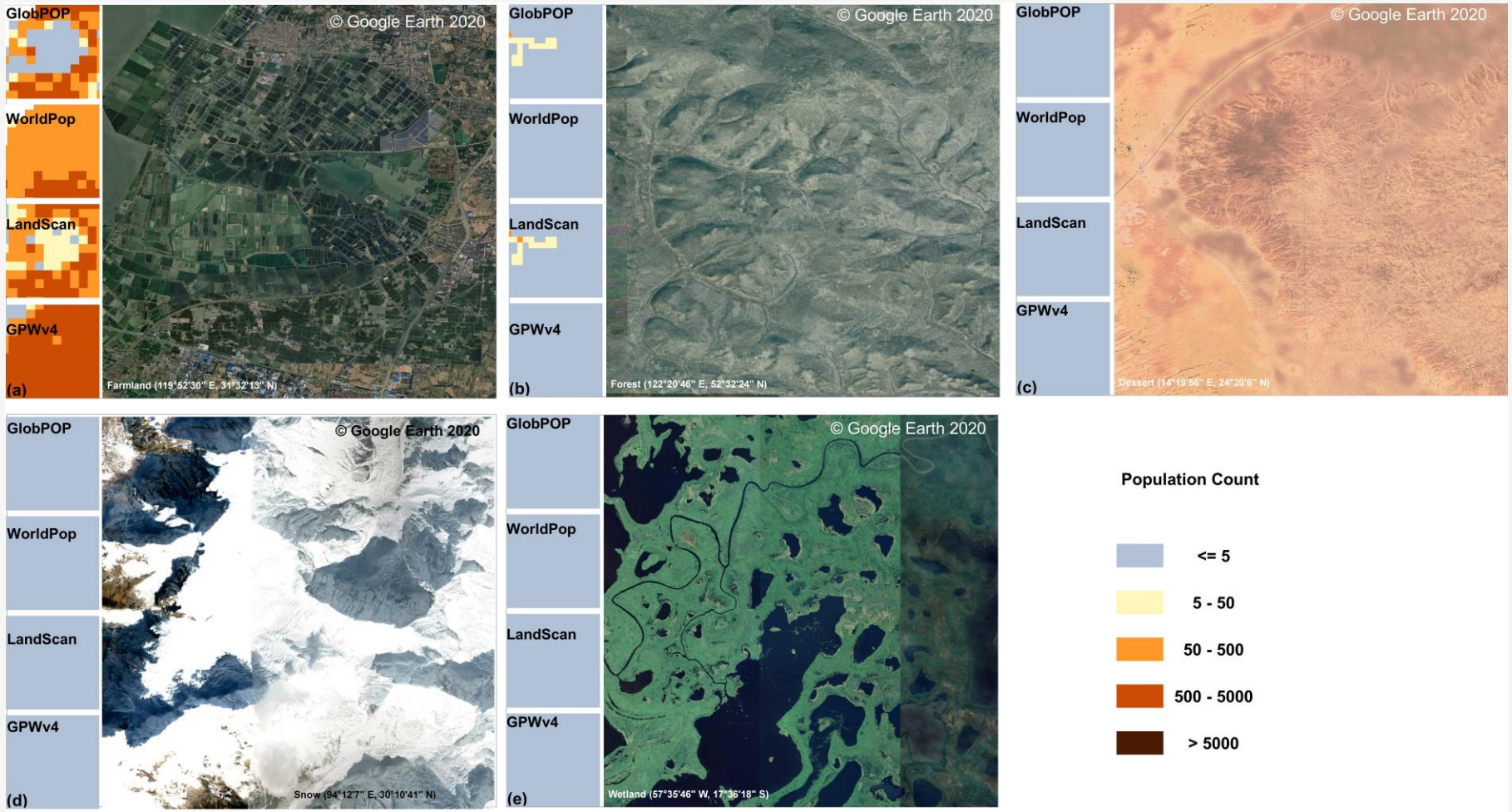


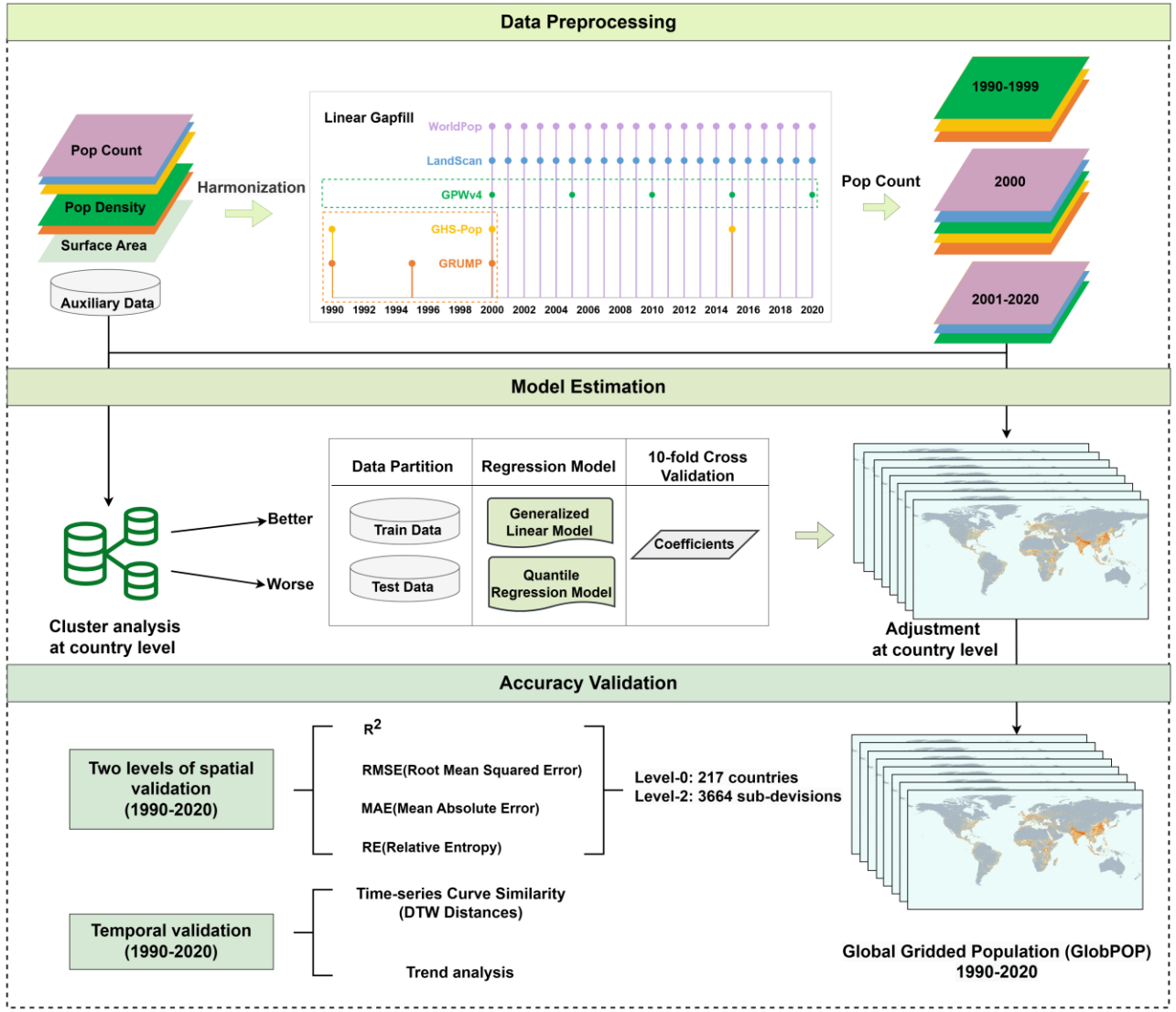
Fig. 10 Examples of population distribution at pixel level and the google earth image in 2020. (a) Farmland in western China; (b)Forest in northern China; (c) The Sahara Desert in Africa; (d) Snow mountain in west-eastern China; (5) Pantanal wetland in South America

? Why visual inspection

There is a **lack of reference data** to perform spatial validation *statistically* at the pixel level

- GlobPOP: outperforms in cropland but remains equivalent in other land cover types

5 Conclusion *Innovations*



Innovation 1: Global Population Data Fusion Framework

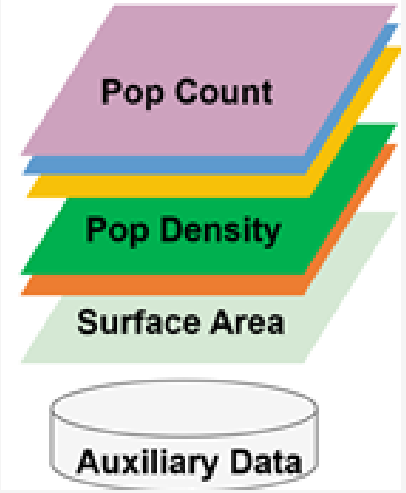
- Introduced a novel data fusion framework that successfully generates continuous global gridded population data (GlobPOP) from 1990 to 2020

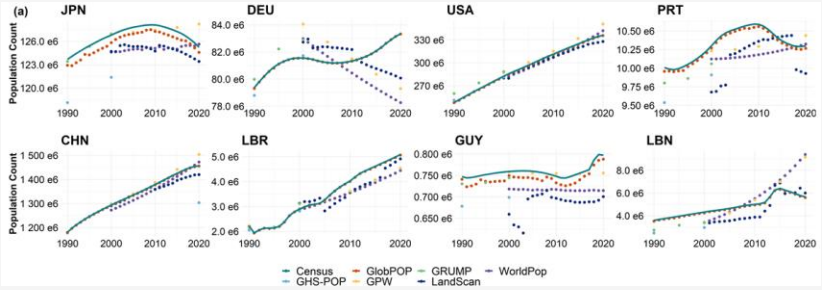
Innovation 2: Temporal and Spatial Integration for Validation

- Conducted in-depth validation at both spatial and temporal scales.
- Provided the accountability distribution map and interactive country population curve for comparison.

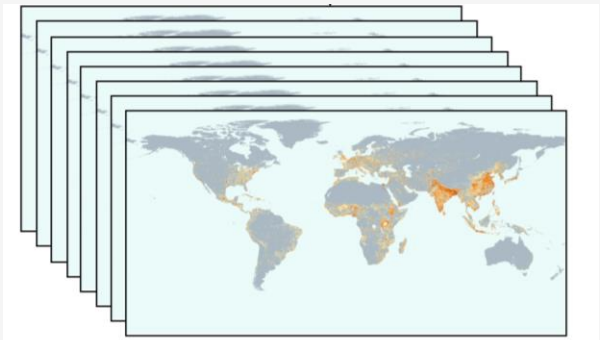
5 Conclusion

Usage notes

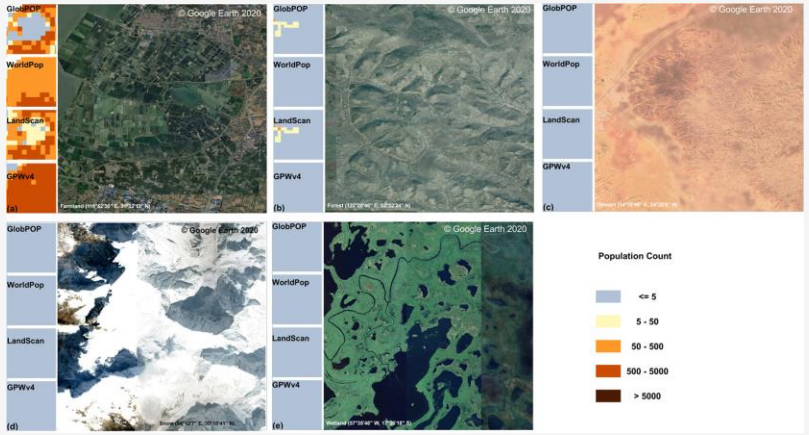




Spatial and Temporal Consistent Performance



Global Gridded Population (GlobPOP)
1990-2020



Researchers and policymakers can leverage the new dataset to conduct time-series analysis of the population and explore the spatial patterns of population development at global, national, and city levels